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Paper 33
Date: April 6, 2022

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

RIMFROST AS,
Petitioner,

v.

AKER BIOMARINE ANTARCTIC AS,
Patent Owner.

IPR2020-01533
Patent 9,816,046 B2

Before ERICA A. FRANKLIN, JON B. TORNQUIST, and MICHAEL A. VALEK, *Administrative Patent Judges.*

TORNQUIST, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. Background and Summary

Rimfrost AS ("Petitioner") filed a Petition (Paper 2, "Pet.") requesting an *inter partes* review of claims 1–19 of U.S. Patent No. 9,816,046 B2 (Ex. 1001, "the '046 patent"). Aker Biomarine Antarctic AS ("Patent Owner") did not file a Preliminary Response to the Petition. Upon review of Petitioner's arguments and evidence, we instituted an *inter partes* review of all claims and grounds asserted in the Petition (Paper 6, "Institution Decision" or "Inst. Dec.").

Patent Owner subsequently filed a Response (Paper 9, "PO Resp."), to which Petitioner filed a Reply (Paper 15, "Pet. Reply"), and Patent Owner filed a Sur-Reply (Paper 18, "Sur-Reply"). With authorization, Patent Owner also filed a paper identifying arguments and evidence in Petitioner's Reply that it considers to be improper (Paper 19), to which Petitioner filed a response (Paper 21).

Petitioner also filed a motion to exclude certain evidence relied upon by Patent Owner (Paper 25), to which Patent Owner filed an opposition (Paper 26), and Petitioner filed a reply (Paper 28).

In support of their respective positions, Petitioner relies upon the declaration and reply declaration of Dr. Stephen J. Tallon (Exs. 1006, 1086), and Patent Owner relies upon the declaration of Dr. Jacek Jaczynski (Ex. 2015).

An oral hearing was held on January 12, 2022, and a transcript of the hearing is included in the record (Paper 32, "Tr.").

B. Real Parties-in-Interest

Petitioner identifies itself, Olympic Holding AS, Emerald Fisheries AS, Rimfrost USA, LLC, Rimfrost New Zealand Limited, and Bioriginal Food and Science Corp. as real parties-in-interest. Pet. 3. Based on various ownership interests, and out of "an abundance of caution," Petitioner also identifies Stig Remøy, SRR Invest AS, Rimfrost Holdings AS, and Omega Protein Corporation as real parties-in-interest. *Id.*

Patent Owner identifies itself as the real party-in-interest in this proceeding. Paper 5, 1.

C. Related Matters

The parties identify as a related matter *Aker Biomarine Antarctic AS v. Olympic Holding AS*, 1:16-CV-00035-LPS-CJB (D. Del.), which involved U.S. Patent Nos. 9,028,877 B2 ("the '877 patent") and 9,078,905 B2 ("the '905 patent"). Pet. 3–4; Paper 5, 1. The parties further identify Investigation No. 337-TA-1019 by the United States International Trade Commission, which involved the '877 and '905 patents, as well as U.S. Patent No. 9,320,765 ("the '765 patent"), U.S. Patent No. 9,375,453 ("the '453 patent"), and U.S. Patent No. 9,072,752 ("the '752 patent"). Pet. 4; Paper 5, 1–2.

The parties also identify the following Board proceedings as related matters:

- IPR2017-00745 and IPR2017-00747, which requested review of the '905 patent (all challenged claims found unpatentable (Ex. 1103), decision affirmed on appeal (Ex. 1154));
- IPR2017-00746 and IPR2017-00748, which requested review of the '877 patent (all challenged claims found unpatentable (Ex. 1104), decision affirmed on appeal (Ex. 1154));

- IPR2018-00295, which requested review of the ’765 patent (all challenged claims found unpatentable (Ex. 1129));
- PGR2018-00033, which requested review of U.S. Patent No. 9,644,170 (institution denied because the challenged patent was not eligible for post grant review);
- IPR2018-01178 and IPR2018-01179, which requested review of the ’453 patent (all challenged claims found unpatentable (Exs. 1157, 1158));
- IPR2018-01730, which requested review of the ’752 patent (all challenged claims found unpatentable (Ex. 1159)); and
- IPR2020-01532, which requested review of U.S. Patent No. 9,644,169 B2 (pending).

Pet. 4–7; Paper 5, 2–4.

D. The ’046 Patent

The ’046 patent discloses extracts from Antarctic krill that include bioactive fatty acids. Ex. 1001, 1:24–25. The ’046 patent explains that krill oil compositions, including compositions having up to 60% w/w phospholipid content and as much as 35% w/w EPA/DHA[1] content, were known in the art. *Id*. at 1:59–62. The ’046 patent further explains that “[k]rill oil compositions have been described as being effective for decreasing cholesterol, inhibiting platelet adhesion, inhibiting artery plaque formation, preventing hypertension, controlling arthritis symptoms, preventing skin cancer, enhancing transdermal transport, reducing the

[1] According to the ’046 patent, “EPA” is 5,8,11,14,17-eicosapentaenoic acid and “DHA” is 4,7,10,13,16,19-docosahexanoic acid. Ex. 1001, 9:13–16.

symptoms of premenstrual symptoms or controlling blood glucose levels in a patient." *Id.* at 1:51–57.

According to the '046 patent, lipases and phospholipases within krill can result in the decomposition of glycerides and phospholipids, such as phosphatidylcholine, during transport of frozen krill from the Southern Ocean to a processing site. *Id.* at 2:11–18, 9:61–63. To avoid the problem of enzymatic decomposition of krill products, the '046 patent describes a method of thermally denaturing the lipases and phospholipases in fresh-caught krill prior to storage and processing. *Id*. at 9:63–10:11, 10:45–50. The '046 patent reports that these denaturing steps allow for the storage of krill material "for from about 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, or 12 months to about 24 to 36 months prior to processing." *Id*. at 10:36–44.

The '046 patent describes an embodiment wherein krill oil is subsequently extracted from the stored krill product using a polar solvent and/or supercritical carbon dioxide. *Id.* at 10:2–4. In Example 7 of the '046 patent, "[k]rill lipids were extracted from krill meal (a food grade powder) using supercritical fluid extraction with co-solvent." *Id.* at 32:15–16.

> Initially, 300 bar pressure, 333°K and 5% ethanol (ethanol:CO_2, w/w) were utilized for 60 minutes in order to remove neutral lipids and astaxanthin from the krill meal. Next, the ethanol content was increased to 23% and the extraction was maintained for 3 hours and 40 minutes. The extract was then evaporated using a falling film evaporator and the resulting krill oil was finally filtered.

Id. at 32:17–24.

In Example 8, krill oil prepared using the same method described in Example 7 was analyzed using ^{31}P NMR to identify and quantify the phospholipids in the oil. *Id.* at 32:48–51. It was determined that "[t]he main polar ether lipids of the krill meal [were] alkylacylphosphatidylcholine (AAPC) at 7-9% of total polar lipids, lyso-alkylacylphosphatidylcholine (LAAPC) at 1% of total polar lipids (TPL), and alkylacylphosphatidyl-ethanolamine (AAPE) at <1% of TPL." *Id.* at 33:13–17.

E. Illustrative Claims

Petitioner challenges claims 1–19 of the '046 patent. Claims 1 and 13 are independent and claims 2–12 depend, directly or indirectly, from claim 1 and claims 14–19 depend from claim 13. Claims 1 and 13 are illustrative of the challenged claims and are reproduced below:

> 1. A method of production of krill oil comprising:
>
> obtaining a krill meal produced by a process comprising treating krill to destroy the activity of lipases and phospholipases naturally present in krill and wherein said krill meal has been stored for period of from 1 to 36 months; and
>
> extracting krill oil from said krill meal that has been stored from 1 to 36 months with a polar solvent to provide a krill oil with greater than 30% phosphatidylcholine w/w of said krill oil and astaxanthin esters.

Ex. 1001, 35:48–57.

> 13. A method of production of *Euphausia superba* krill oil comprising:
>
> a) obtaining a *Euphausia superba* krill meal produced by a process comprising treating *Euphausia superba* to destroy the activity of lipases and phospholipases naturally present in *Euphausia superba* and wherein said *Euphausia superba* krill meal has been stored from 1 to 36 months; and
>
> b) extracting *Euphausia superba* oil from said krill meal that has been stored from 1 to 36 months with a polar solvent to provide a *Euphausia superba* krill oil comprising greater than 30% phosphatidylcholine w/w of said *Euphausia superba* krill oil, less than 3% free fatty acids w/w of said *Euphausia superba* krill oil, and at least 100 mg/kg astaxanthin esters.

Id. at 36:42–56.

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–19 would have been unpatentable on the following grounds:

Claims Challenged	35 U.S.C. §[2]	References/Basis
1–10	103(a)	Breivik II,[3] Yoshitomi,[4] Budziński,[5] Fricke,[6] Bottino II,[7] Sampalis I[8]
11, 12	103(a)	Breivik II, Yoshitomi, Budziński, Fricke, Bottino II, Randolph,[9]
13–19	103(a)	Breivik II, Yoshitomi, Budziński, Fricke, Bottino II, Randolph, Sampalis I

Pet. 11.

[2] The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) ("AIA"), amended 35 U.S.C. §§ 102 and 103. Because the '046 patent has an effective filing date before the March 16, 2013, effective date of the applicable AIA amendments, we refer to the pre-AIA versions of 35 U.S.C. §§ 102 and 103.

[3] Breivik, WO 2008/060163 A1, published May 22, 2008 (Ex. 1037). Breivik claims priority to U.S. Provisional Application No. 60/859,289, filed November 16, 2006. Ex. 1037, code [30].

[4] Yoshitomi et al., US 2003/0113432 A1, published June 19, 2003 (Ex. 1033).

[5] Budziński, E., et al., *Possibilities of processing and marketing of products made from Antarctic krill*, FAO Fish.Tech. Pap., (268):46 (1985) (Ex. 1008).

[6] Fricke et al., *Lipid, Sterol and Fatty Acid Composition of Antarctic Krill (Euphausia superba Dana)*, 19(11) LIPIDS 821–827 (1984) (Ex. 1010).

[7] Bottino, *Lipid Composition of Two Species of Antarctic Krill: Euphausia superba and E. crystallorophias*, 50B COMP. BIOCHEM. PHYSIOL. 479–484 (1975) (Ex. 1038).

[8] Sampalis et al., *Evaluation of the Effects of Neptune Krill Oil™ on the Management of Premenstrual Syndrome and Dysmenorrhea*, 8(2) ALT. MED. REV. 171–179 (2003) (Ex. 1012).

[9] Randolph et al., US 2005/0058728 A1, published Mar. 17, 2005 (Ex. 1011).

II. ANALYSIS

A. Legal Standards

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *See KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) if in the record, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. Level of Ordinary Skill in the Art

Petitioner provides the following definition of the ordinarily skilled artisan

> A [person of ordinary skill in the art] would have held an advanced degree in marine sciences, biochemistry, organic (especially lipid) chemistry, chemical or process engineering, or associated sciences with complementary understanding, either through education or experience, of organic chemistry and in particular lipid chemistry, chemical or process engineering, marine biology, nutrition, or associated sciences; and knowledge of or experience in the field of extraction. In addition, a [person of ordinary skill in the art] would have had at least five years applied experience.

Pet. 9–10 (citing Ex. 1006 ¶¶ 36–37).

Neither Patent Owner nor Dr. Jaczynski disputes Petitioner's definition of a person of ordinary skill in the art. Ex. 2015 ¶ 8 ("For purposes of this proceeding, I will accept [Petitioner's] definition of a [person of ordinary skill in the art].").

Upon review of the '046 patent and the prior art of record, we find Petitioner's uncontested definition reasonable and supported by the evidence of record. Accordingly, we adopt this definition of one of ordinary skill in the art for purposes of this Decision.

C. Claim Construction

Petitioner provides proposed claim constructions for the terms "krill oil," "polar solvent," "astaxanthin ester," "krill meal," and "destroy the activity of lipases and phospholipases." Pet. 33–38. With respect to "krill meal," Petitioner contends this term should be construed to mean "processed krill with reduced water content from which krill oil can be extracted." *Id.* at 37.

Patent Owner contends the term "destroy the activity of lipases and phospholipases" should be given its plain and ordinary meaning. PO Resp. 11. With respect to "krill meal," Patent Owner contends Petitioner's proposed construction of "krill meal" is overbroad and ignores the ordinary and customary meaning of a "meal," which is a "powder formed by particle size reduction of a starting material." *Id.* at 8 (citing Ex. 2015 ¶¶ 13–15). Thus, Patent Owner construes "krill meal" to mean "a krill powder resulting from the processing of krill." *Id.* at 11.

Upon review of the parties' arguments and supporting evidence, we determine that no terms of the '046 patent require construction in order to resolve the disputes in this proceeding. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803

(Fed. Cir. 1999) ("[O]nly those terms need be construed that are in controversy, and only to the extent necessary to resolve the controversy.")).

D. Collateral Estoppel

Petitioner contends that, with the exception of Budziński, each of the "references relied on by Petitioner was evaluated and applied by the Board in IPRs involving Patent Owner's" related patents. Pet. 39. Petitioner further contends that in two prior IPRs the Board analyzed similar subject matter to that claimed in the '046 patent and found the claims unpatentable. *Id.* at 40. "Given the materially identical claim limitations and common references," Petitioner contends "the Board's factual findings and conclusions of law regarding, *inter alia*, the teachings of the prior art, the motivation to combine those teachings, and claim construction in the prior Final Written Decisions should be given preclusive effect." *Id.*

As is set forth and explained below, the parties' primary dispute in this proceeding is whether one of ordinary skill in the art would have found it obvious to extract krill oil from a denatured krill meal that has been stored from 1 to 36 months. Petitioner concedes that this precise issue was not considered in the prior IPRs, but contends collateral estoppel still applies because the "inclusion of an intuitive and common sense storage limitation does not materially alter the question of the '046 patent's unpatentability." Pet. Reply 6–7.

"Collateral estoppel protects a party from having to litigate issues that have been fully and fairly tried in a previous action and adversely resolved against a party-opponent." *Ohio Willow Wood Co. v. Alps South, LLC*, 735 F.3d 1333, 1342 (Fed. Cir. 2013). In the context of patent claims, collateral estoppel applies "[i]f the differences between the unadjudicated patent claims and adjudicated patent claims do not materially alter the question of invalidity." *Id.*

Petitioner points to no prior decision of the Board where the issue of whether it would have been obvious to denature a krill product and then store it for 1 to 36 months before extraction was at issue. Moreover, to demonstrate that the added limitation would have been obvious, Petitioner relies heavily on the disclosures of Budziński, which was not at issue or analyzed in the prior proceedings. Pet. 39. Accordingly, Petitioner has not demonstrated that collateral estoppel applies to any issue in this proceeding.

E. Reliance on Breivik II

Patent Owner contends the invention recited in the claims of the '046 patent was conceived and reduced to practice prior to the earliest possible priority date of Breivik II. PO Resp. 20. And, because each ground asserted in the Petition relies on Breivik II, Patent Owner contends the Petition must be denied. *Id.* at 38.

Petitioner asserts that Patent Owner has failed to antedate Breivik II. Pet. Reply 7.

For the reasons set forth below, we determine that Petitioner has shown by a preponderance of the evidence that challenged claims 1–19 would have been obvious over Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I and/or Randolph, when considered in light of the undisputed knowledge of one of ordinary skill in the art. Thus, we need not decide whether Patent Owner's evidence of conception and reduction to practice is sufficient to antedate Breivik II with respect to these claims, as we do not rely on it as a prior art reference in this proceeding.

F. Claims 1–19 over Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I

Petitioner contends that the subject matter of claims 1–10 would have been obvious over the combined disclosures of Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I. Pet. 41–57.

1. Yoshitomi

Yoshitomi discloses that krill enzymes cause krill to lose freshness in a short period of time after being hauled on-board a ship and that these enzymes may be disabled or inactivated either by quickly freezing the krill below -40º C or by heating the krill above 80º C. Ex. 1033 ¶¶ 5–6.

Yoshitomi explains that various denatured krill products were known in the art, including "boiled krill products which are heated and then preserved in a frozen condition" and krill meals that are heated, dried, and then "preserved at the normal temperature." *Id.* ¶ 7. To improve upon these known products, Yoshitomi describes chopping krill material into pieces and then placing the pieces into a heating and drying machine. *Id.* ¶ 37. Yoshitomi notes that this process results in "a dried powdery and granular krill product" that "contains all components of krill and in which lipid degradation is sufficiently prevented" without using an antioxidant. *Id*. ¶¶ 2, 9, 22 (noting that the heating process results in a krill meal wherein "the proteolytic enzymes originally contained in krill materials are perfectly disabled").

Yoshitomi provides experimental evidence of the stability of its krill meal. *Id.* ¶¶ 43–45, Table 3. In one example, Yoshitomi stores two denatured krill meals at 37º C, one with and one without additional antioxidants. *Id.* ¶ 43. Until the end of the first month of storage, there was no significant difference in lipids between the two krill meals. *Id.*¶ 44.

After two months of storage, oxidation proceeded "slightly faster in the group added with no anti-oxidant than the group added with the anti-oxidant, but a significant difference was not found." *Id.* ¶¶ 44, 46 (noting that in the disclosed krill product "lipid degradation can be prevented satisfactorily without using any anti-oxidant").

2. Budziński

Budziński, published in 1985, "presents the current state of possible commercial-scale uses of krill-processing technologies," including processing technologies for products for "human consumption, animal feed, and industrial purposes." Ex. 1008, Abstr. Budziński explains that the "main difficulty in krill processing" is caused by proteolytic enzymes, such as lipases. *Id.* at 6.[10] According to Budziński, lipases "cause the decomposition of phospholipids and, to a lesser degree, triglycerides," and result in an increase in free fatty acids in krill oil. *Id.* at 6–7. To avoid enzyme degradation of krill products, Budziński explains that storage time for raw krill products "should be as short as possible" and that vessels "must be properly equipped for processing" krill as soon as possible after capture. *Id.* at 9, 20 ("Efforts should be made to regulate catches so that the time before processing is as short as possible."), 25 ("Due to its technological properties, the raw materials should be processed as soon as possible after capture. The only way to meet this requirement is to install processing facilities on-board the vessel."). Budziński discloses that the optimum temperature for heating krill is believed to be 80–85 °C and that a properly

[10] Unless otherwise noted, citations to prior art references are to the original page numbers of the documents, not to the page numbers added by Petitioner in the lower right corner of the documents.

processed krill meal is "stable for 13 months of storage without addition of antioxidants." *Id.* at 20–22.

3. Fricke

Fricke reports the total lipid content and composition for krill samples caught in the Antarctic Ocean in December 1977 and March 1981. Ex. 1010, Abstr., 822. Fricke reports that "[k]rill samples of 5 kg were quick-frozen and stored at -35 C until analyzed" and that "lipid extraction was performed according to Folch et al." *Id.* at 821. The samples analyzed in Fricke contained approximately 33.3% +/- 0.5% w/w (1977 sample) and 40.4% +/- 0.1% w/w (1981 sample) triacylglycerols and 16.1 (1977 sample) and 8.5 (1981 sample) percent free fatty acids. *Id*. at 822.

Fricke notes that samples from the 1977 sample contained free fatty acid levels that were about twice that of the 1981 sample. *Id.* at 822. In contrast, samples of the same haul that had been cooked on board "immediately after hauling," showed a free fatty acid content that "was much lower, ranging from 1% to 3% of total lipids." *Id.* at 822–23.

4. Bottino II

Bottino II characterizes the lipids of two species of Antarctic krill, *Euphasia superba* and *Euphasia crystallorophias*. Ex. 1038, 479. Bottino II explains that "when one refers to Antarctic krill, one generally means *Euphausia superba*, which is the most abundant and far better known species of krill in the Antarctic Oceans." *Id.*

Bottino II describes a process wherein euphausiids were collected and, once on board the ship, the samples rapidly sorted by hand and extracted with a "chloroform:methanol (2:1, v/v) mixture." *Id.* The fatty acid content of the krill product was then determined by gas-liquid chromatography. *Id.* at 480.

Table 2 of Bottino II is reproduced below.

Table 2. The lipids of Antarctic krill

	E. superba		*E. crystallorophias*	
	Station 8 (1)*	Station 11 (2)	Station 13 (4)	Station 16 (2)
	weight %			
Waxes	—	—	44±10‡	20±1
Steroid esters	—	—	2±3	27±9
Triglycerides	8	36±6	—	—
Diglycerides	17	4±5	—	4±1
Complex lipids	54	58±14	53±8	42±8
PC†		48	46	
PE†		8	6	
Lyso PC		1	1	
PG†		1	—	
Unknown§	21	2±22	1±2	7±1

* Number of determinations in parentheses.
† PC, Phosphatidylcholine; PE, phosphatidylethanolamine; PG, phosphatidylglycerol.
‡ Weight per cent plus or minus the standard deviation.
§ R_f between those of triglycerides and diglycerides. The recovered amount of this fraction was too small for further characterization.

Ex. 1038, 481 (Table 2). Table 2 reports the identity and amount of each lipid present in the *E. superba* and *E. crystallorophias* samples obtained from various locations (i.e., stations) (analyzed as a weight percent of total lipids). *Id.* at 480–481. As shown in Table 2, the phosphatidylcholine content of krill obtained from Stations 11 and 13 was 48% and 46%, respectively, and the lysophosphatidylcholine (i.e., Lyso PC) content was 1% in both samples. *Id.* at 481 (Table 2), 483.

5. *Sampalis I*

Sampalis I discloses the use of Neptune Krill Oil, or NKO, and notes that this product is extracted from *Euphausia superba* and contains "various potent antioxidants including vitamins A and E, astaxanthin," and at least one novel flavonoid. Ex. 1012, 174. Sampalis I also describes the health benefits of Neptune Krill Oil and reports that two soft gels of the product taken daily "can significantly reduce the physical and emotional symptoms related to premenstrual syndrome." *Id.* at Abstr.

6. Analysis of Claim 1

a) "obtaining krill meal produced by a process comprising treating the krill to destroy the activity of lipases and phospholipases"

Petitioner contends that obtaining a krill meal through a process of treating krill to destroy the activity of lipases and phospholipases was well known in the art, as evidenced by Yoshitomi, Budziński, and Fricke. Pet. 42–44.

Yoshitomi discloses heating krill to denature and "perfectly" disable "the proteolytic enzymes originally contained in krill materials." Ex. 1033 ¶ 22. The result is "a dried powdery and granular krill product that contains all components of krill" and can be stored for more than two months without significant degradation of krill lipids. *Id.* ¶¶ 9, 22, 24, 42–46; Pet. 43.

Budziński discloses that "very active lipases" present in krill can cause decomposition of phospholipids and increase the level of free fatty acids, and that these enzymes can be "fully denatured" using heat. Ex. 1008, 12, 26; Pet. 43. Budziński further discloses that the resulting krill meal is "stable for 13 months of storage without addition of antioxidants." Ex. 1008, 21–22.

Fricke discloses that krill that are heated immediately after hauling on-board a ship have a "much lower" level of free fatty acids, "ranging from 1% to 3% of total lipids." Ex. 1010, 822–23.

Patent Owner does not expressly dispute that this limitation was taught or suggested in the prior art.

Given these disclosures, we find that Yoshitomi, Budziński, and Fricke each teach or suggest treating krill to destroy the activity of lipases and phospholipases, and Yoshitomi and Budziński further disclose obtaining a stable, denatured krill meal. Ex. 1006 ¶¶ 171–172, 288–296, 417–419.

b) "wherein said krill meal has been stored for a period of from 1 to 36 months"

Petitioner contends that Budziński and Fricke teach or suggest storing a denatured krill meal for a period of from 1 to 36 months. Pet. 44.

Budziński expressly discloses denaturing krill using heat, and explains that "[e]xtensive investigations" on various types of krill meal "have shown that they are stable for 13 months of storage without addition of antioxidants." Ex. 1008, 21–22. Fricke discloses that a krill product that was "cooked on board immediately after hauling" and then stored before extraction had a "much lower" level of free fatty acids than products that were not cooked before storage. Ex. 1010, 823.

Patent Owner does not expressly dispute that it was known in the art that a denatured krill meal could be stored for between 1 and 36 months.

Upon review of the record, we find that Budziński discloses a denatured krill meal that was successfully stored for at least 13 months and that Fricke teaches or suggests that denatured krill products may be stored for an extended period of time prior to extraction. As such, these references teach or suggest storing a denatured krill meal "for a period of from 1 to 36 months."[11]

[11] Yoshitomi describes a denatured krill meal that was stored for at least two months without evidence of significant oxidation. Ex. 1033 ¶¶ 43–46; Ex. 1006 ¶¶ 296, 417; Pet. 44 n.7 (Petitioner noting the "good preservation ability" of Yoshitomi's krill meal).

c) *"extracting krill oil from said krill meal that has been stored from 1 to 36 months with a polar solvent"*

Petitioner contends that extracting krill oil using a polar solvent was known in the art, as described in Fricke, Bottino II, and Budziński. Pet. 45; Ex. 1006 ¶¶ 424–428, 431, 463.

Budziński discloses storing krill for up to 13 months and extracting krill oil using "various organic solvents" and provides tables showing the chemical composition of krill meals. Ex. 1008, 21–24, Tables 11–13. Fricke discloses cooking krill to form a denatured product, storing the product for a period of time, and then extracting lipids from this product using a polar solvent. Ex. 1010, 821–823 (noting that krill samples were frozen and lipids then subsequently extracted using the method of Folch et al.); Ex. 1017, 509 (Folch describing the extraction lipids from animal tissue using a 2:1 Chloroform-methanol mixture); Ex. 1006 ¶ 300 (Dr. Tallon explaining that Folch uses a polar solvent for extraction). And Bottino II discloses catching krill and then extracting lipids using a polar solvent. Ex. 1038, 479–480; Ex. 1006 ¶ 428.

Patent Owner asserts that Petitioner's evidence with respect to this claim limitation is insufficient because neither Budziński, Fricke, nor Bottino II discloses using a polar solvent to extract krill oil from a krill meal that has been stored from 1 to 36 months. PO Resp. 38–39. With respect to Budziński, Patent Owner concedes that this reference discloses a "stable" krill meal that can be stored for up to 13 months without antioxidants, and that krill oil can be extracted using various organic solvents, but contends that these constitute "two separate, unrelated statements" that are not "a disclosure of the claim element of extracting krill oil from a krill meal after storage of the krill meal for from 1 to 36 months." *Id.* at 39.

Patent Owner further contends that Fricke discloses extraction from "cooked krill," not a krill meal, and Fricke "contains no teaching as to the storage time for the cooked krill, how the cooked krill was extracted, or lipid classes other than free fatty acids that may or may not have been present in the cooked krill." *Id.* at 41.

With respect to Bottino II, Patent Owner contends this reference contains no disclosure of extracting krill oil from a krill meal, "much less krill meal that has been stored for from 1 to 36 months." *Id.* at 44.

According to Patent Owner, because "none of the references relied on by Petitioner teach . . . 'extracting krill oil from said krill meal that has been stored from 1 to 36 months with a polar solvent,'" Petitioner has failed to carry its burden of proving that the combined references teach this claim limitation. *Id.* at 44–45.

When addressing the question of obviousness, we consider what the disclosures of the prior art, as a whole, would have taught or suggested to one of ordinary skill in the art. *See Bradium Techs. v. Iancu*, 923 F.3d 1032, 1050 (Fed. Cir. 2019). We do not limit our analysis to what each reference discloses in isolation. *Id.* ("A finding of obviousness, however, cannot be overcome 'by attacking references individually where the rejection is based upon the teachings of a combination of references.'"); *see Fleming v. Cirrus Design Corp.*, No. 2021-1561, 2022 WL 710549, at *6 (Fed. Cir. March 10, 2022) (explaining that "it is sufficient that a person of ordinary skill in the art would have been motivated to combine the prior art in a way such that *the combination discloses the claim limitations*") (emphasis added).

Upon review of the record as a whole, we find that it was known in the art that krill oil could be extracted from krill products using a polar solvent, including from denatured krill products stored for a period of time before extraction. Ex. 1008, 24; Ex. 1010, 821–823; Ex. 1038, 479–480; Ex. 1006 ¶ 424; Ex. 1086 ¶ 26; Ex. 2001, 14; Ex. 2015 ¶ 27. The prior art of record also demonstrates that denatured krill meals could be successfully stored for between 1 and 36 months. Ex. 1008, 21–22 (Budziński disclosing that "[e]xtensive investigations" have shown that denatured krill meals are "stable for 13 months of storage without addition of antioxidants"); *see also* Ex. 1033 ¶¶ 24, 44, Tables 2, 3 (Yoshitomi disclosing that denatured krill meal can be stored at normal temperatures due to its low water content and that krill lipids were not significantly oxidized after two months of storage). Finally, Dr. Tallon persuasively testifies that it was understood in the art that krill oil, including known astaxanthin esters and phosphatidylcholine constituents, could be successfully extracted from krill products. Ex. 1006 ¶¶ 423–424, 429, 431, 434; *see also* Ex. 1008, 24 (Budziński disclosing the extraction of krill oil using "various organic solvents"). Thus, Petitioner persuasively explains why the prior art of record, in combination, teaches or suggests "extracting krill oil from said krill meal that has been stored from 1 to 36 months with a polar solvent," as recited in claim 1.

d) "to provide a krill oil with greater than 30% phosphatidylcholine w/w of said krill oil and astaxanthin esters"

Petitioner contends it was understood in the art that both phosphatidylcholine and astaxanthin esters are naturally present in krill and that both components would be extracted from a krill meal using "conventional extraction techniques and known polar solvents." Pet. 18, 45–

46 (citing Ex. 1006 ¶¶ 429–437 (Dr. Tallon noting that Yoshitomi, Budziński, and Sampalis I all disclose that astaxanthin is naturally present in krill oil)); *see also id.* at 27 (noting that Sampalis I discloses a krill oil containing astaxanthin). Petitioner further contends that Bottino II discloses extracting a krill oil with 48% phosphatidylcholine. *Id.* at 46 (citing Ex. 1038, 481).

Patent Owner does not contest that Bottino II discloses a krill oil with 48% phosphatidylcholine, that it was understood in the art that phosphatidylcholine and astaxanthin esters are naturally present in krill, or that it was understood in the art that krill oil could be extracted using conventional techniques, such as polar solvents. *See generally* PO Resp. 38–53.

Upon review of the prior art of record and the testimony of both Dr. Tallon and Dr. Jaczynski, we agree with Petitioner that it was understood in the art that both phosphatidylcholine and astaxanthin esters are naturally present in krill and would be extracted using polar solvents. Pet. 18, 27, 45–46 (citing Ex. 1038, Table 2 (Bottino II disclosing a krill oil with 48% phosphatidylcholine); Ex. 1006 ¶¶ 55 (Dr. Tallon testifying that it was understood in the art before 2007 that astaxanthin was commercially important and known to be naturally present in krill), 66 (Dr. Tallon noting the sale of commercial krill oils containing >1500 mg/kg astaxanthin esters), 69, 74, 434 (Dr. Tallon testifying that the krill meal of Budziński contained astaxanthin that would have been extracted using a polar solvent), 433–437; Ex. 1170, 62:10–63:10 (Dr. Jaczynski testifying during cross-examination that before November of 2006 one of ordinary skill in the art would have known that astaxanthin esters are natural components of krill oil that can be extracted in desired amounts by known methods); Ex. 2015 ¶¶ 50–51 (Dr.

Jaczynski conceding that astaxanthin is a natural component of krill); *see also* Ex. 1008, 21 (Budziński disclosing the presence of astaxanthin and its esters in its denatured, stable krill meal); Ex. 1033 ¶ 34 (Yoshitomi disclosing that astaxanthin is present in krill); Ex. 1011 ¶¶ 39, 44 (Randolph disclosing that krill oils obtained using "[c]onventional oil producing techniques" typically contain "between about 0.5 mg and about 50 mg of an astaxanthin ingredient"); Ex. 1012, 174 (Sampalis I disclosing a krill oil containing astaxanthin). We also find, in view of the evidence, that Bottino II discloses a krill oil with greater than 30% phosphatidylcholine. Pet. 46 (citing Ex. 1038, Table 2 (Bottino II disclosing a krill oil with 48% phosphatidylcholine); Ex. 1006 ¶¶ 414–437).

Accordingly, Petitioner persuasively demonstrates that Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I, in combination with the knowledge of one of ordinary skill in the art, teach or suggest "to provide a krill oil with greater than 30% phosphatidylcholine w/w of said krill oil and astaxanthin esters," as recited in claim 1.

7. Reason to Combine

Petitioner contends that Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I are each directed to the same field of endeavor and that one of ordinary skill in the art would have sought to combine their disclosures to arrive at the methods set forth in claims 1–10 of the '046 patent in view of the known health benefits of krill oil. Pet. 51.

Petitioner argues that one of ordinary skill in the art would have recognized the health benefits of krill oil and its constituent components, including phosphatidylcholine and astaxanthin, in view of the disclosures of Sampalis I, Bottino II, and Randolph. *Id.* at 54–56 (citing Ex. 1012, 178; Ex. 1020, 421; Ex. 1011; Ex. 1100, 185, 191).

To obtain a useful krill oil containing phosphatidylcholine and astaxanthin, Petitioner argues that one of ordinary skill in the art would have sought to denature krill lipases and phospholipases in fresh caught krill in order "to prevent spoilage and the formation of undesirable free fatty acids and degraded phospholipids, such as lysophospholipids," as disclosed in Yoshitomi, Budziński, and Fricke. *Id.* at 51 (citing Ex. 1006 ¶¶ 75, 78, 96, 128, 161, 244, 249, 258, 302, 303, 462).

After harvesting and processing kill at sea to obtain a denatured krill meal, Petitioner contends one of ordinary skill in the art would have sought to store the denatured krill meal for between 1 and 36 months in order to perform land-based solvent extraction. *Id.* at 53. Petitioner argues that land-based extraction would have been beneficial because it would avoid the expense of installing and operating extraction facilities on a ship, avoid the hazards associated with using flammable polar solvents at sea, and allow for intermittent scheduling of oil extraction to satisfy fluctuating demand for freshly extracted krill oil. *Id.* at 53–54 (citing Ex. 1006 ¶¶ 54, 75, 78, 171, 172, 175, 177, 420–423).

Once the krill oil is extracted, Petitioner contends one of ordinary skill in the art would have encapsulated this product to provide a "convenient dosage form," as disclosed in Sampalis I. *Id.* at 56 (citing Ex. 1012, 174; Ex. 1011 ¶ 52; Ex. 1006 ¶¶ 65, 200, 380).

Patent Owner contends Petitioner's reasons to combine the recited references are not persuasive because Breivik II[12] and Budziński instruct that

[12] Because Patent Owner relies on various disclosures of Breivik II as potentially counselling against combining the references to arrive at the claimed subject matter, we consider Breivik II here only in this context. We do not, however, consider Breivik II or Petitioner's arguments with

extraction should be performed on fresh krill, not a denatured krill meal that has been stored for an extended period of time. PO Resp. 51–53. Patent Owner further contends that Petitioner's reasons to combine the identified prior art references are "in direct conflict" with the teachings of Breivik II and Budziński, as well as admissions made by Dr. Tallon during his deposition. *Id.* at 45–51.

a) Alleged Teaching Away

Patent Owner argues that the meaning of "stable" would "depend entirely on the context and what the krill meal was being used for," and the krill meal of Budziński was stable only with respect to use as an animal feed. PO Resp. 52; Ex. 2019, 26:24–27:8. Given the uncertainty regarding the stability of the krill meal of Budziński, Patent Owner contends one of ordinary skill in the art would have sought to extract krill oil from fresh krill, not a krill meal that was stored for between 1 and 36 months. PO Resp. 51–52.

Budziński also teaches that "since extraction costs are high and special installations are needed, it is unrealistic to expect production of krill oil in practice." PO Resp. 39 (quoting Ex. 1008, 24). Patent Owner asserts that, if anything, this disclosure "teaches away from extracting oil from krill." *Id.*

Patent Owner's arguments with respect to Budziński are unavailing. First, although stability may depend on context, Dr. Tallon persuasively explains that the presence of astaxanthin in the krill meal of Budziński demonstrates both the quality of its cooking and drying process and that the

respect to this reference in determining whether the prior art teaches or suggests every limitation of the challenged claims.

resulting krill meal was stable with respect to both enzymatic breakdown and oxidation. Ex. 1086 ¶ 51 (Dr. Tallon testifying that the presence of astaxanthin demonstrated that the krill meal was stable with respect to oxidation and enzyme activity). Thus, we find that one of ordinary skill in the art would have understood that Budziński's krill meal was stable with respect to both oxidation and enzymatic breakdown and was suitable for extraction to produce krill oil. *Id.*; Ex. 1006 ¶¶ 422 (asserting that krill oil could be obtained from a stable, denatured krill meal "using any of the known extraction methods"), 434.

Second, although Budziński asserts that krill oil production is unrealistic due to the high extraction costs, it also notes the high potential value of krill oil products, and subsequent references disclose successfully extracting krill oil for use in pharmaceuticals or nutritional products. Ex. 1008, 24 (Budziński noting that "krill oil could be a valuable raw material for oil, cosmetic and pharmaceutical industries"); Ex. 1009, 24:1–17 (Catchpole); Ex. 1011 ¶¶ 39–40, 44 (Randolph); Ex. 1012, 174 ("Neptune Krill Oil™ (NKO™) is a natural health product extracted from Antarctic krill . . . ."). As such, we agree with Petitioner that due to subsequent advances in the art that occurred prior to the earliest possible priority date of the '046 patent, Budziński would not have taught away from, or counseled against, extracting krill oil from krill products. Pet. Reply 20–21.

Breivik II discloses "a process for preparing a substantially total lipid fraction from fresh krill without using organic solvents like acetone." Ex. 1037, 3:29–31. In this process, fresh krill are optionally pre-treated with heat to inactivate enzymes that would decompose lipids, and then extracted using carbon dioxide/ethanol. *Id.* at 3:33–4:6. According to Breivik II, these disclosed processes are "expected" to result in "substantially less hydrolysed and/or oxidized lipids than lipids produced by conventional processes" and in "less deterioration of the krill lipid antioxidants." *Id.* at 3:35–4:3.

Patent Owner contends that one of ordinary skill in the art considering Breivik II's disclosures would have understood that extraction of fresh krill is preferable over stored krill products, and that extraction should be performed on the fishing vessel as opposed to on land. PO Resp. 53.
On this record, we agree with Patent Owner that Breivik II teaches or suggests a successful method of immediately processing and extracting fresh caught krill on-board a fishing vessel. We also agree that Breivik II suggests that its disclosed method may have advantages over conventional krill oil extraction processes. The disclosed advantages to immediately denaturing and extracting krill oil discussed in Breivik II, however, are only "expected," not definitive, and we are directed to no disclosure in Breivik II to suggest that extraction from a stored, denatured krill meal was unlikely to result in a useful krill oil. *See In re Fulton*, 391 F.3d 1195, 1200 (Fed. Cir. 2004) (noting that the case law "does not require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation for the current invention"); *In re Gurley*, 27 F.3d 551, 553 (Fed. Cir. 1994) (explaining that a reference teaches away when it suggest that a particular course of action "is unlikely to be productive of the result sought by the applicant," and that "[a] known or obvious composition does not become patentable simply because it has been described as somewhat inferior to some other product of the same use").

Moreover, although, as noted in Breivik II, oxidation and enzymatic hydrolysis are of particular concern during krill processing procedures, Petitioner and Dr. Tallon persuasively demonstrate that the denatured and "stable" krill meal of Budziński was not subject to significant enzymatic hydrolysis or deterioration of krill lipid antioxidants, such as astaxanthin. Ex. 1006 ¶¶ 172, 421, 434; Ex. 1086 ¶¶ 50–51; *see also* Ex. 1010, 822–823 (Fricke disclosing storing denatured krill products for a period of time before extraction with a polar solvent). As such, we are not persuaded that Breivik II would have taught away from, or otherwise discouraged, one of ordinary skill in the art from extracting krill oil from a denatured and stable krill meal that has been stored from between 1 and 36 months.

Patent Owner further argues that, consistent with Breivik II's disclosures, Budziński instructs that krill should be "processed as soon as possible after capture" and that "[t]he only way to meet this requirement is to install processing facilities on-board the vessel." PO Resp. 51 (citing Ex. 1008, 25). The required processing of Budziński, however, is to denature the problematic krill enzymes, not to extract krill oil. Ex. 1008, 25. And, once the krill products are denatured, Petitioner persuasively demonstrates that krill oil extraction could be successfully performed either on-board a vessel or on land after a storage period of up to 13 months.

In view of the foregoing, we find that Breivik II and Budziński do not teach away from, or counsel against, extracting krill oil from a krill meal that had been denatured and stored for 1 to 36 months.

b) Flammability Concerns and Benefits of Land-Based Extractions

Patent Owner argues that Petitioner's "basis for combining the references to arrive at the claimed method . . . relies on conclusions reached by its expert, Dr. Tallon, regarding processing on ships and in particular to the use of flammable solvents on ships." PO Resp. 45. Patent Owner asserts, however, that Dr. Tallon admitted during his deposition that "there was really no problem with using flammable solvents on a ship." *Id.* at 47 (citing Ex. 2019, 49:18–25). Given this admission, as well as Breivik II's disclosure of successfully extracting krill oil on a fishing vessel, Patent Owner contends "[t]here is no basis to conclude that extraction would preferably be done on land on stored material due to safety concerns," as asserted by Petitioner and Dr. Tallon. *Id.* at 48.

Contrary to Patent Owner's assertion, Petitioner need not demonstrate that one of ordinary skill in the art would have preferred to extract krill oil on land, as opposed to on a ship. *See Fulton*, 391 F.3d at 1200. Rather, Petitioner is required to explain why one of ordinary skill in the art, considering all the factors involved with ship-based and land-based extractions, would have seen a benefit or reason to extract krill oil in a land-based facility. *KSR*, 550 U.S. at 418 (looking to "whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue").

With respect to flammability concerns, we agree with Patent Owner that the record indicates that the risks of using flammable solvents on a fishing vessel could be successfully managed. Ex. 2019, 49:18–25. We also credit the testimony of Dr. Tallon, however, that the risks and costs of using flammable solvents on a vessel are still relevant. Ex. 1086 ¶¶ 105–106. Generally supporting Dr. Tallon's testimony, Budziński states that "[t]he use of flammable solvents for extraction makes production at sea unrealistic, and it seems that this option is not very promising at present." Ex. 1008, 18. Likewise, Grantham notes that solvent extraction on-board a vessel could be accomplished, "but there are substantial technical and safety problems" with this approach. Ex. 1032, 28. Breivik II's subsequent disclosures, as well as

Dr. Tallon's cross-examination testimony suggesting that concerns in using flammable solvents can be managed, does not establish that the flammability concerns expressed in Budziński and Grantham were no longer relevant. Ex. 1086 ¶¶ 103–109. Accordingly, we find that one of ordinary skill in the art would have considered flammability concerns when deciding whether to use ship-based or land-based krill oil extraction methods.

Petitioner and Dr. Tallon also identify additional benefits to land-based extractions, such as avoiding the costs of implementing and running extraction facilities on-board a ship, and the ability to extract fresh krill oil to meet current product demand. Pet. 53–54; PO Resp. 50. Patent Owner does not explain why these asserted benefits are incorrect, but asserts that Dr. Tallon's testimony is based on "speculative conclusions." PO Resp. 50. We see nothing speculative, however, regarding the additional costs required to install krill extraction facilities on a ship and to carry additional fuel to run the extraction process on each krill fishing vessel, or in the benefit of being able to meet fluctuating demand for freshly extracted krill oil. Pet. 53.

Patent Owner further argues that Dr. Tallon's testimony is contrary to his reasons for storing a krill meal for 1 to 36 months. PO Resp. 50. In particular, Patent Owner contends Dr. Tallon testified that "there really is no difference between facilities installed on board a ship and on land." *Id.* at 50–51 (citing Ex. 2019, 59:14–23). The fact that krill oil extraction facilities are similar on a ship and on land, however, does not address the cost of installing and running such facilities on-board each ship, as opposed to on land. Nor does this argument address the additional benefit of land-based extraction facilities in being able to provide freshly extracted krill oil to meet customer demand. Pet. 53. As such, Dr. Tallon's deposition testimony does not appear to conflict with his proffered reasons to combine the recited prior art references.

In view of the foregoing, we find that one of ordinary skill in the art would have understood that both ship-based and land-based extractions of krill oil were known in the art, both methods would result in a useful krill oil, and that there were benefits to extracting krill oil in a land-based facility. Ex. 1006 ¶¶ 54, 75, 78, 171–172, 175, 177, 420–423; *KSR*, 550 U.S. at 421 ("When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp."). We further find that one of ordinary skill in the art would have: (1) understood the health benefits of krill oil, including its naturally occurring phosphatidylcholine and astaxanthin constituents, (2) sought to denature krill to prevent the activity of enzymes, (3) sought to use a polar solvent to extract krill oil from a denatured krill meal that was stored between 1 and 36 months, and (4) sought to encapsulate krill oil to provide a convenient dosage form. Accordingly, we find persuasive Petitioner's explanation as to why one of ordinary skill in the art would have been motivated to combine the teachings of Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I, to arrive at the subject matter of independent claim 1.

8. Conclusion with Respect to the Combination of Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I

Petitioner persuasively identifies where Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I, in combination with the knowledge of one of ordinary skill in the art, teach or suggest every limitation of claim 1 of the '046 patent. Petitioner also persuasively explains why one of ordinary skill in the art would have sought to combine the disclosures of Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I to arrive at the subject matter of claim 1 with a reasonable expectation of success. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claim 1 would have been obvious over Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I.

9. Dependent Claims 2–10

Claims 2–10 each depend, directly or indirectly, from claim 1. Ex. 1001, 35:58–36:32. Petitioner contends that Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I, in combination, teach or suggest every limitation of these dependent claims, when considered in light of the knowledge of one of ordinary skill in the art. Pet. 46–51. In particular, the record shows that: Bottino II and Fricke both disclose krill oil extracts with less than 3% free fatty acids (claim 2) (*id.* at 46–47 (citing Ex. 1038, 481; Ex. 1010, 823)); Bottino II discloses a krill oil extract with less than 2% lysophosphatidylcholine (claim 3) (*id.* at 47 (citing Ex. 1038, 481; Ex. 1006 ¶¶ 440–441)); Yoshitomi discloses grinding krill prior to destroying the activity of lipases and phospholipases naturally present in the krill (claim 4) (*id.* at 47–48 (citing Ex. 1033 ¶¶ 9, 22, 32, 37, 41, 48–50, Fig. 2; Ex. 1006 ¶ 442)); Yoshitomi, Budziński, and Fricke each disclose using heat to destroy the activity of krill lipases and phospholipases (claim 5) (*id.* at 48 (citing Ex. 1033, Abstract; Ex. 1008, 24; Ex. 1010, 822–823; Ex. 1006 ¶¶ 443–446)); using both chemicals and heat to denature or destroy krill lipases and phospholipases was well known and "standard" in the art[13]

[13] Claims 6, 7, 18, and 19 additionally require that the krill be treated with chemicals or a combination of heat and chemicals to denature the krill enzymes. The Petition relies on Breivik II as teaching those limitations. Pet. 48–49, 65–66 (citing Ex. 1037, 3:33–4:2; Ex. 1006 ¶¶ 447–450, 519–

(claims 6 and 7) (*id.* at 48–49; PO Resp. 24–25, 30 (Patent Owner arguing for conception and reduction to practice that any differences between its antedating evidence and the claims "are obvious"); Ex. 1037, 3:33–4:6; Ex. 1086 ¶ 26 (Dr. Tallon testifying that a person of ordinary skill in the art understood that "heat and/or chemicals" could be used to destroy lipases and phospholipases in krill material); Ex. 1170, 56:21–58:4 (Dr. Jaczynski testifying that it was well known in the art that chemicals, or heat and chemicals, were standard methods to destroy krill lipases and phospholipases); Ex. 2001, 16 (named inventor, Dr. Tilseth, testifying to show conception and reduction to practice of this claim limitation that "[h]eat and chemical treatment is a standard way to destroy the activity of enzymes")); Sampalis I discloses krill oil in an encapsulated dosage form (claim 8) (Pet. 49–50 (citing Ex. 1012, 174); Bottino II, Yoshitomi, Fricke, and Budziński disclose using *Euphausia superba* as a source of krill oil (claim 9) (*id.* at 50 (citing Ex. 1038, 479; Ex. 1033 ¶ 29; Ex. 1010, 221;

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522). Patent Owner contends Breivik II is not prior art with respect to claims 6, 7, 18, and 19 because those claims were conceived and reduced to practice prior to the earliest priority date of Breivik II. PO Resp. 24–25, 30. However, Patent Owner does not identify any evidence that the named inventors conceived of or reduced to practice a method of denaturing krill enzymes using chemicals, or chemicals and heat. Instead, Patent Owner asserts that such methods were well known in the art and would have been "obvious" to one of ordinary skill in the art. *Id.* at 24–25, 30 (citing *In re Spiller*, 500 F.2d 1170 (CCPA 1974); *In re Stryker*, 435 F.2d 1340 (CCPA 1971)). Given Patent Owner's assertions, we need not decide the antedation issue, as we hold Patent Owner to its admission that the use of chemicals and heat to denature krill enzymes was well known in the art. And, when the knowledge of one of ordinary skill in the art is undisputed, this knowledge "must be consulted when considering whether a claimed invention would have been obvious." *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013); *see Fleming*, 2022 WL 710549, at *6.

Ex. 1008, 1, 3)); and Bottino II discloses a krill oil with at least 40% phosphatidylcholine (claim 10) (*id.* at 50–51 (citing Ex. 1038, Table 2)).

Patent Owner does not address, much less contest, Petitioner's arguments and evidence related to dependent claims 2–10, beyond its arguments directed to independent claim 1, addressed above. *See, e.g.*, PO Resp. 53.

Upon review of Petitioner's arguments and supporting evidence, as well as the admissions of Patent Owner, Dr. Jaczynski, and Dr. Tilseth regarding the knowledge of one of ordinary skill in the art, we find that Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I, in combination with the knowledge and skill in the art, teach or suggest every limitation of dependent claims 2–10. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claims 2–10 would have been obvious over the disclosures of Yoshitomi, Budziński, Fricke, Bottino II, and Sampalis I.

G. Claims 11 and 12 over Yoshitomi, Budziński, Fricke, Bottino II, and Randolph

Claim 11 depends from claim 1 and further requires "wherein said astaxanthin esters are present in the krill oil at in an amount of at least 100 mg/kg." Ex. 1001, 36:36–38. Claim 12 also depends from claim 1 and further requires "wherein said astaxanthin esters are present in the krill oil at in an amount of at least 200 mg/kg." *Id.* at 36:39–41.

Petitioner contends the subject matter of claims 11 and 12 would have been obvious over the combined disclosures of Yoshitomi, Budziński, Fricke, Bottino II, and Randolph. Pet. 57–59.

1. Randolph

Randolph discloses various compositions for modulating an inflammatory or immunomodulatory response, including krill oil. Ex. 1011 ¶ 8. The krill oil of Randolph may be obtained from any member of the *Euphausia* family, including *Euphausia superba*, and typically contains "between about 300 mg and about 3000 mg of a krill oil ingredient." *Id.* ¶¶ 39–40. Randolph further teaches that the composition can contain any amount of an astaxanthin ingredient, including at least about 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, or 90 percent astaxanthin, typically resulting in between about 0.5 mg and about 50 mg of astaxanthin in the product. *Id.* ¶ 44.

Randolph explains that the ingredients of the composition can be processed into various "delivery systems," including capsules, soft gel capsules, tablets, gel tabs, lozenges, strips, granules, powders, concentrates, solutions, lotions, creams or suspensions. *Id.* ¶¶ 46, 52. Randolph further explains that a soft gel capsule of the composition can be manufactured to include about 300 mg of krill oil. *Id.* ¶ 52.

2. Analysis

Petitioner argues that using the "lower value for astaxanthin (0.5 mg) and the upper value (0.003 kg) for krill oil, Randolph discloses krill oil compositions having at least 167 mg/kg of astaxanthin (0.5 mg/0.003 kg), which is equivalent to 158 mg/kg astaxanthin esters." Pet. 57–58 (citing Ex. 1011 ¶¶ 40, 44; Ex. 1006 ¶¶ 192, 469–470). Petitioner further argues that by simply increasing Randolph's lower astaxanthin limit and/or decreasing its upper krill oil limit, one of ordinary skill in the art would have arrived at a krill oil composition with more than 200 mg/kg astaxanthin esters. *Id.* at 58.

Petitioner contends one of ordinary skill in the art would have sought to combine the teachings or Randolph with those of Yoshitomi, Budziński, Fricke, and Bottino II in view of Randolph's teachings that astaxanthin is useful in treating a variety of afflictions, as well as its teaching of krill oil compositions with greater than 100 mg/kg astaxanthin esters. *Id.* at 59.

Patent Owner contends Petitioner's arguments with respect to Yoshitomi, Budziński, Fricke, Bottino II, and Randolph fail because claims 11 and 12 depend from claim 1 and Randolph does not cure the deficiencies identified by Patent Owner with respect to claim 1. PO Resp. 54–55 (asserting that Randolph includes "no details . . . on how the krill oil was produced or any suggestion to extract krill oil from a krill meal that has been stored for from 1 to 36 months"). Patent Owner does not address, however, Petitioner's specific arguments with respect to Randolph and its astaxanthin content. *See, e.g., id.*

Upon review of the record as a whole, we find that Yoshitomi, Budziński, Fricke, Bottino II, and Randolph, in combination, teach or suggest every limitation of claims 11 and 12. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claims 11 and 12 would have been obvious over Yoshitomi, Budziński, Fricke, Bottino II, and Randolph.

H. Claims 13–19 over Budziński, Fricke, Yoshitomi, Bottino II, Randolph, and Sampalis I

Claim 13 is independent and differs from independent claim 1 in requiring the use of *Euphausia superba* krill to produce the claimed krill meal and in expressly requiring "less than 3% free fatty acids w/w" and "at least 100 mg/kg astaxanthin esters." Ex. 1001, 36:42–56; PO Resp. 56. As noted above with respect to dependent claims 2, 9, and 11, Petitioner demonstrates that use of *Euphausia superba* was well known in the art, obtaining less than 3% free fatty acids w/w is disclosed in both Fricke and Bottino II, and Randolph teaches or suggests a krill oil with "at least 100 mg/kg astaxanthin esters." Pet. 46–47, 50, 57–58, 60. As such, Petitioner demonstrates that Yoshitomi, Budziński, Fricke, Bottino II, Randolph, and Sampalis I teach or suggest every limitation of independent claim 13. *See id.* at 60–64 (Petitioner addressing every limitation of claim 13).

Petitioner also argues that every limitation of dependent claims 14–19 is disclosed in one or more of Yoshitomi, Budziński, Fricke, Bottino II, Randolph, and Sampalis I. Pet. 64–66. In particular, Petitioner contends that both Sampalis I and Randolph disclose encapsulating krill oil (claim 14) (*id.* at 64 (citing Ex. 1006 ¶ 451; Ex. 1012, 174; Ex. 1011 ¶ 52)); Bottino II discloses a *Euphausia superba* extract with 48% phosphatidylcholine (claim 15) (*id.* at 46, 64 (citing Ex. 1038, Table 2; Ex. 1006 ¶¶ 459, 511)); Yoshitomi discloses grinding, chopping, or crushing *Euphausia superba* and then heating this product to produce a "dried powdery and granular" product or meal (claim 16) (*id.* at 47–48, 65 (citing Ex. 1033 ¶¶ 9, 22, 32, 37, 41, 48–50, Figure 2; Ex. 1006 ¶¶ 442, 514)); Yoshitomi, Budziński, and Fricke each disclose using heat to denature or destroy the activity of krill lipases and phospholipases (claim 17) (*id.* at 48, 65 (citing Ex. 1033, Abstract; Ex. 1008, 24; Ex. 1010, 822–823; Ex. 1006 ¶¶ 443–446, 515-518)); and Patent Owner asserts that the use of chemicals and heat to denature or destroy krill lipases and phospholipases was well known and standard in the art (claims 18 and 19) (*id.* at 65–66 (citing Ex. 1037, 3:33–4:6); PO Resp. 24–25, 30; Ex. 1006 ¶ 286; Ex. 1086 ¶ 26 (Dr. Tallon testifying that a person of ordinary skill in the art understood that "heat and/or chemicals" could be used to destroy lipases and phospholipases in krill material); Ex. 1170, 56:21–58:4 (Dr. Jaczynski testifying that use of chemicals, or heat and chemicals, was a standard method to destroy krill lipases and phospholipases); Ex. 2001, 16 (Dr. Tilseth testifying on the issue of conception and reduction to practice that "[h]eat and chemical treatment is a standard way to destroy the activity of enzymes")).

Patent Owner does not address Petitioner's specific arguments with respect to claims 13–19, but asserts that these claims would not have been obvious for the same reasons set forth above with respect to claim 1. PO Resp. 56–57.

Upon review of the parties' arguments and the evidence of record, including the knowledge of one of ordinary skill in the art, we find that Petitioner persuasively identifies where Yoshitomi, Budziński, Fricke, Bottino II, Randolph, and Sampalis I teach or suggest every limitation of claims 13–19. Thus, Petitioner demonstrates by a preponderance of the evidence that claims 13–19 would have been obvious over Yoshitomi, Budziński, Fricke, Bottino II, Randolph, and Sampalis I.

III. REPLY ARGUMENTS

Patent Owner sought authorization to file a motion to strike portions of Petitioner's Reply. We denied this request, but permitted Patent Owner to file a three-page paper identifying any evidence or argument in the Reply that it considered to be improper. Paper 19, 2. We also authorized Petitioner to file a three-page response to Patent Owner's submission. *Id.*

A. "Some Months" Argument and Prior Board Decision

The Petition asserts that Fricke teaches or suggests storage of processed krill prior to solvent extraction. Pet. 44 (citing Ex. 1010, 822–823; Ex. 1006 ¶ 423). In response, Patent Owner argued that Fricke "contains no teaching as to the storage time of the cooked krill."

PO Resp. 41–42. Petitioner replied to this argument by noting that in IPR2017-00746 the Board previously found that the samples of Fricke were extracted after being stored for "some months." Pet. Reply 21–22 (citing Ex. 1104, IPR2017-00746, Paper No. 23, at 28–29; Ex. 2006; Ex. 1086 ¶¶ 68–71).

Patent Owner contends the argument that Fricke's denatured krill samples were stored for "some months," as well as Petitioner's citation to Exhibit 1160, which was first filed with Petitioner's Reply, constitute improper new arguments and evidence "intended to supplement Petitioner's arguments" in the Petition. Paper 19, 2. Patent Owner further contends that the Board's decision in IPR2017-00746 was not cited in the Petition and is improperly intended to supplement Petitioner's arguments in the Petition. *Id.* at 2–3.

Because we do not rely on either the Board's discussion of Fricke in the Final Written Decision in IPR2017-00746 or Petitioner's argument that the krill of Fricke were stored for "some months," Patent Owner's objections are moot.

IV. PETITIONER'S MOTION TO EXCLUDE

Petitioner moves to exclude Exhibits 2003, 2010, and 2013, as well as paragraphs 7, 8, 12–14, and 17 of the Declaration of Dr. Tilseth (Ex. 2001) and paragraph 27 of the Declaration of Dr. Jaczynski (Exhibit 2015). Paper 25, 1.

Each of the documents and paragraphs identified by Petitioner relates to Patent Owner's attempt to swear behind Breivik II to remove it as a prior art reference in this proceeding. *Id.* at 2–13. As we do not reach this issue in this Decision, Petitioner's Motion to Exclude is dismissed as moot.

V. CONCLUSION[14]

In consideration of the above, we find that Petitioner demonstrates by a preponderance of the evidence that claims 1–19 are unpatentable as obvious over the combined teachings of the prior art of record, when considered in light of the knowledge of one of ordinary skill in the art.

In summary:

Claims	**35 U.S.C. §**	**Reference(s)/Basis[15]**	**Claims Shown Unpatentable**	**Claims Not Shown Unpatentable**
1–10	103(a)	Breivik II, Yoshitomi, Budziński, Fricke, Bottino II, Sampalis I	1–10	
11, 12	103(a)	Breivik II, Yoshitomi, Budziński, Fricke, Bottino II, Randolph	11, 12	
13–19	103(a)	Breivik II, Yoshitomi, Budziński, Fricke, Bottino II, Randolph, Sampalis I	13–19	
Overall			1–19	

[14] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

[15] As noted above, we have considered Petitioner's grounds without relying on Breivik II. We do, however, consider whether Breivik II would teach away from or counsel against the combinations of prior art asserted in the Petition, as asserted by Patent Owner.

Outcome				

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–19 of the ’046 patent are unpatentable;

FURTHER ORDERED that Petitioner’s Motion to Exclude is dismissed as moot; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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